

SKILL - 19

Code:

```
GNU nano 7.2 skill19_signals.c
#define _POSIX_C_SOURCE 200809L

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <errno.h>
#include <string.h>

volatile sig_atomic_t child_pid = 0;

void handle_signal(int sig)
{
    int saved_errno = errno;
    pid_t pid = (pid_t)child_pid;

    if (sig == SIGINT) {
        const char msg[] =
            "\nParent: SIGINT received. Forwarding to child.\n";
        write(STDOUT_FILENO, msg, sizeof(msg) - 1);
    }

    if (pid > 0)
        kill(pid, sig);
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ ls
hi skill19_signals skill19_signals.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ sudo apt update
```

```
249 packages can be upgraded. Run 'apt list --upgradable' to see them.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ ^C
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 249 not upgraded.
```

```
0 upgraded, 0 newly installed, 0 to remove and 249 not upgraded.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ ./skill19_signals
SIGINT handler registered successfully.
Parent PID: 2560
Press Ctrl+C to interrupt the child.
Child process started. PID: 2561
Child working... 1/30
Child working... 2/30
Child working... 3/30
Child working... 4/30
Child working... 5/30
Child working... 6/30
Child working... 7/30
Child working... 8/30
Child working... 9/30
Child working... 10/30
Child working... 11/30
Child working... 12/30
Child working... 13/30
Child working... 14/30
Child working... 15/30
Child working... 16/30
Child working... 17/30
^C
Parent: SIGINT received. Forwarding to child.
Child terminated by signal 2.
Parent process is still running safely.
```

```
Child working... 11/30
Child working... 12/30
Child working... 13/30
Child working... 14/30
Child working... 15/30
Child working... 16/30
Child working... 17/30
Child working... 18/30
Child working... 19/30
Child working... 20/30
Child working... 21/30
Child working... 22/30
Child working... 23/30
Child working... 24/30
Child working... 25/30
Child working... 26/30
Child working... 27/30
Child working... 28/30
Child working... 29/30
Child working... 30/30
Child completed normally.
Child exited with status 0.
Parent process is still running safely.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$ nano skill19_signals.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-19$
```

